Akshay Thakare

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Experience Summary

- Highly skilled and motivated Cloud Data Engineer with experience in designing, building, and optimizing large-scale data pipelines and infrastructure. Proficient in utilizing big data technologies such as Hadoop, Apache Kafka, Spark, and Airflow. Experienced in cloud platforms like AWS and proficient in SQL, NoSQL databases, Linux, and Python
- Strong understanding of Spark Architecture including Spark Core, Spark-SQL, DataFrame, Datasets, RDD, and performance tuning with partitioning, caching, and broadcast joins.
- Previously drove cloud cost optimization at Tata Technologies by analyzing job loads, execution timing, and infrastructure usage, achieving 35% reduction in cloud spend.
- Hands-on experience in Data Warehouses (Hive, Snowflake) - Managed tables, partitions, bucketing, micro-partitions, Hive UDFs, and optimization techniques.
- Worked with SQL, HQL & NoSQL databases such as MySQL, PostgreSQL, Apache HBASE, and ELK
- Designed, scheduled, and monitored Azure Data Factory pipelines. Reported ETL pipelines between internal and external data stores and optimizing data processing.
- Currently leading SAS-to-PySpark migration at Capgemini, converting 50,000+ lines of legacy ETL code to scalable Databricks pipelines with Delta Lake integration

Key Skills

- **Big Data Technologies:** Apache Spark, PySpark, Hadoop, EMR, Kafka, Kafka Connect, Databricks, Delta Lake
- **Data Processing & Orchestration:** Apache Airflow, Oozie, Cron Jobs, CI/CD Pipelines
- **Programming & Scripting**: Python (PySpark), SQL, HQL, SAS (migration expertise), Shell Scripting (Unix/Linux), REST APIs, FastAPI, Flask
- **Databases:** SQL (MySQL, PostgreSQL), NoSQL (HBase, Elasticsearch, MongoDB)
- **Data Warehouses:** Hive, Snowflake (Partitions, Bucketing, UDFs, Optimization Techniques, Micro-Partitions)
- **Data Engineering:** Data Modeling, Schema Evolution, SCD Type 1/2, Star Schema, Data Lineage, Metadata-Driven Frameworks
- **Cloud Services:** AWS (EC2, S3, IAM, RDS, Glue, EMR, Athena, Lambda, CloudWatch), Azure (ADF, Databricks, Blob Storage), Azure DevOps, Delta Lake
- **Cloud Optimization**: Job Load Analysis, Execution Timing Profiling, Infrastructure Right-Sizing, Cost Optimization, Performance Tuning
- **Containerization & DevOps**: Docker (Airflow, ELK, MongoDB, Kafka, Jenkins, MySQL, Ranger), Kubernetes
- **Data Governance:** Apache Ranger
- **Monitoring & Logging**: ELK Stack (Elasticsearch, Logstash, Kibana)
- **Gen-AI:** LLaMA, Ollama, Streamlit, Prompt Engineering, Natural Language to SQL/DSL
- **Tools & Platforms:** Git, Azure DevOps, Visual Studio Code, JIRA, Agile/Scrum

Experience

- Currently working as **Consultant (B2)** in **Capgemini** India Pvt. Ltd, Mumbai since **May - 2025**.
- Worked as **Sr. Developer** in **Tata Technologies Limited**, Thane, Mumbai from **Dec 2022 - May 2025**.

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Projects

1.Type of Industry:	**Automotive**
Project Description:	Developed a scalable end-to-end data pipeline for processing large-scale streaming (1.5 TB daily) and transactional data using Apache Kafka, Spark, Python, and Aws services.
Role/Title:	Sr. Developer
Project Duration:	March 2024 – April 2025
Responsibilities:	• Designed real-time data ingestion pipeline from Kafka to MongoDB and MongoDB to ELK Index by using PySpark. • Enabled real-time data insights, enhancing fleet performance monitoring and decision-making. • Improved data processing efficiency, reducing ETL execution time by 40%. • Scaled data pipeline to handle high-volume streaming data with minimal latency.
Operating Systems:	Amazon Linux-2, Red-Hat
Software / Languages:	AWS, Hadoop, Python, Kafka, SQL, Spark, ELK, Hive, MongoDB, Airflow, Docker, Linux
Special Tools:	Apache Kafka.

2.Type of Industry:	**Automotive**
Project Description:	Developed a real-time fault detection and analysis system using Kafka, Python and ELK Stack to enable proactive customer engagement for vehicle fault resolution.
Role/Title:	Sr. Developer
Project Duration:	Jan 2023 – Feb 2024
Responsibilities:	• Processed fault signals and vehicle location data from Kafka topics • Performed real-time joins for enriched fault analysis. • Implemented real-time alerting and indexing using Elasticsearch. • Improved operational efficiency by enabling real-time fault tracking. • Reduced downtime and enhanced customer satisfaction through proactive issue resolution. • Provided vendors with data-driven insights for better customer service and predictive maintenance.

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Operating Systems:	Amazon Linux-2, Red-Hat
Software / Languages:	Apache Kafka, PySpark, Elasticsearch (ELK Stack), MongoDB, Python, SQL
Special Tools:	Apache Kafka

3. Type of Industry **Finance**

Project Description:	Leading end-to-end SAS-to-PySpark migration for enterprise finance client, converting 50,000+ lines of legacy SAS ETL code to scalable PySpark/Databricks pipelines with Delta Lake, Snowflake, and automated validation frameworks for financial data processing and reporting workflows
Role/Title:	Consultant (Data Engineer)
Project Duration:	May 2025 - March 2026
Responsibilities:	• Spearheading conversion of 50,000+ lines of SAS ETL code to PySpark, translating complex SAS procedures (PROC SQL, DATA steps, macros) into optimized Spark transformations. • Building automated row-by-row validation frameworks comparing SAS and PySpark outputs, ensuring zero data discrepancy. • Developing reusable Python-based migration templates and utility libraries, accelerating workstream onboarding by 30%. • Implementing data lineage tracking, audit logging, and schema evolution strategies for regulatory compliance. • Collaborating with business stakeholders to translate complex SAS business logic into optimized PySpark transformations. • Targeting 60% pipeline runtime reduction and significant infrastructure cost savings by migrating to cloud-based Databricks.
Operating Systems:	Linux/Unix, SAS, Windows
Software / Languages:	PySpark, Databricks, Snowflake, Delta Lake, SAS, Python, SQL, REST APIs, Apache Airflow, AWS S3
Special Tools:	Databricks, Delta Lake, Apache Airflow

Training, Certifications & Affiliations

- Running Containers on Amazon Elastic Kubernetes Service (Amazon EKS)
- Snowflake[A-Z] Zero to Hero Masterclass(Core+SQL+API+Visual)

Important Links:

- Akshay Thakare – Medium
- apache-airflow-dragdrop-plugin · PyPI
- GitHub

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